

Math 7 — End of Q1 Test · ANSWER KEY

End-of-Q1 Content · 25 questions

Teacher key. Student answers may use equivalent formats (fractions, decimals, with or without units).

1. Convert 3.8 km to mm.

Answer: 3,800,000 mm

2. Convert 450 mg to grams.

Answer: 0.45 g

3. Convert 2.4 L to mL.

Answer: 2400 mL

4. Convert 6200 cm to km.

Answer: 0.062 km

Percent Error Formula (use for questions 5–7):

Percent error = $|\text{measured} - \text{actual}| \div \text{actual} \times 100\%$

5. A student measures a 50 g mass as 45 g. Percent error?

Answer: $|45 - 50| \div 50 \times 100\% = 10\%$

6. A 16 cm pencil is estimated to be 12 cm. Percent error?

Answer: $|12 - 16| \div 16 \times 100\% = 25\%$

7. Scale A reads a 200 kg piano as 190 kg. Scale B reads a 10 kg dog as 8 kg. Which scale is more accurate?

Answer: Scale A (A: $10 \div 200 = 5\%$ error; B: $2 \div 10 = 20\%$ error)

8. Write 6.1×10^{-3} in standard form.

Answer: 0.0061

9. Write 58,000,000 in scientific notation.

Answer: 5.8×10^7

10. $(3 \times 10^4)(3 \times 10^5)$. Write in scientific notation.

Answer: 9×10^9

11. $(8 \times 10^7) \div (2 \times 10^3)$. Write in scientific notation.

Answer: 4×10^4

12. $(3.2 \times 10^5) + (4.5 \times 10^5)$. Write in scientific notation.

Answer: 7.7×10^5

13. $(5 \times 10^4) - (3 \times 10^3)$. Write in scientific notation.

Answer: 4.7×10^4 ($50,000 - 3,000 = 47,000$)

14. $(5 \times 10^3)(2 \times 10^2)$. Write in correct scientific notation.

Answer: $10 \times 10^5 = 1.0 \times 10^6$ (the first number must be at least 1 and less than 10)

15. $(6 \times 10^4) + (7 \times 10^4)$. Write in correct scientific notation.

Answer: $13 \times 10^4 = 1.3 \times 10^5$

16. Is the relationship in the table proportional? (Yes/No) Why?

x	y
0	1
1	5
2	9
3	13
4	17

Answer: No. Why: y/x is not constant ($5/1 = 5$, $9/2 = 4.5$, $13/3 \approx 4.33$), and the graph does not pass through the origin — when $x = 0$, $y = 1$. ($y = 4x + 1$)

17. Find k , the constant of proportionality, for $(5, 35)$. Use $k = y/x$.

Answer: $k = 35/5 = 7$

18. $k = y/x = 8$. If $x = 6$, find y .

Answer: $y = 8 \times 6 = 48$

19. Use $k = y/x$ to find k for the table.

x	y
2	8
3	12
5	20
7	28
10	40

Answer: $k = 4$ ($8/2 = 12/3 = 20/5 = 28/7 = 40/10 = 4$)

20. 180 mi in 3 hrs. mph?

Answer: 60 mph

21. $1/2$ mi in $1/4$ hr. mph?

Answer: 2 mph

22. 6 lb costs \$15. Cost of 1 lb?

Answer: \$2.50

23. 3 cups per 18 cookies. Cups for 54 cookies?

Answer: 9 cups

24. Scale 1 in = 6 ft. Drawing is 7 in. Actual length?

Answer: 42 ft

25. Map 1 cm : 3 km. 12 cm on the map. Actual distance?

Answer: 36 km